GOKULAKANNAN R C

Cloud & DevOps Engineer | AWS | Infrastructure Automation & CI/CD

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PROFESSIONAL SUMMARY

Cloud & DevOps Engineer with nearly 4 years of full-time experience designing and operating AWS infrastructure for SaaS, IoT, and AI applications, with additional Azure deployment experience. Hands-on with CI/CD, Docker, ECS/Fargate, Lambda, Python automation, and Terraform provisioning for selected resources. Delivered approximately 45% AWS cost savings and reduced release time from about 4 hours to 45-50 minutes through infrastructure right-sizing and deployment automation.

TECHNICAL SKILLS

Cloud (AWS): EC2, Lightsail, ECS/Fargate, ECR, Lambda, API Gateway, S3, RDS (MySQL/PostgreSQL), VPC, IAM, KMS, WAF, CloudFront, Route 53, ACM, SES, EventBridge, CloudWatch, Cognito, Parameter Store, Secrets Manager, Elastic Beanstalk, ALB, Auto Scaling

Cloud (Azure): Front Door (CDN), VM Scale Sets, Azure Postgres Flexible Server, Blob Storage, Azure Container Registry, Entra ID, Key Vault, Log Analytics, Virtual Machines, Networking

Infrastructure as Code: Terraform modules for selected AWS resources on HarvestEye and Twellr.ai

CI/CD & Deployment: Bitbucket Pipelines, Jenkins, GitHub Actions, health-checked deployments, rollback

Containers: Docker, Amazon ECS/Fargate, ECR, multi-stage builds, image-size optimization

Monitoring & Logging: AWS CloudWatch, Azure Log Analytics, application logging, usage metrics, Google Analytics

Security & IAM: Least-privilege IAM policy design, KMS key rotation, WAF, SSL/TLS, OAuth2/OIDC (Cognito, Entra ID), VPC security groups, VPN

Scripting & Automation: Python (AWS automation, ETL, webhooks, reporting pipelines, CLI utilities), Bash

Databases: MySQL, PostgreSQL, Amazon RDS, Azure PostgreSQL Flexible Server; schema optimization and data archival

Web Servers & OS: Linux administration, Nginx, Apache, PM2

Version Control: Git, Bitbucket, GitHub

PROFESSIONAL EXPERIENCE

DevOps & Cloud Engineer *January 2023 – Present*

B-Sec Technologies Private Limited — Trichy, India

HarvestEye Portal — Agricultural IoT & Analytics Platform | *AWS: EC2, Lightsail, S3, RDS, ECS/ECR, Lambda, API Gateway, SES, EventBridge, CloudWatch, CloudFront, VPC, Route53, ACM, KMS, WAF, Parameter Store*

- Designed and operated AWS production infrastructure for an Angular frontend on CloudFront/S3, a Django backend on Lightsail/Apache, RDS MySQL, and a dedicated ETL server processing IoT field-device data from S3.
- Provisioned selected AWS resources using Terraform modules, replacing manual setup with repeatable Infrastructure as Code (IaC).
- Built reporting and processing workloads on Lambda and ECS/Fargate using Docker images in ECR; deployed AI/ML services on EC2 and used Jenkins to build memory-intensive PyTorch/YOLO and Detectron2 images.
- Implemented multi-tenant API authentication using IAM SigV4 and database-stored API tokens; automated KMS key-rotation workflows with EventBridge-triggered Lambda functions.
- Configured API Gateway authentication, API keys, throttling, and CORS; automated client report delivery with SES and scheduled processing pipelines with EventBridge.
- Restricted application, database, and ETL traffic using VPC security groups; implemented least-privilege IAM policies, KMS rotation workflows, and WAF rules to strengthen access controls and web-application protection.
- Automated deployments across Lightsail, EC2, ECR, and S3 using Bitbucket Pipelines, including health checks and CloudFront cache invalidation; reduced Docker image size through multi-stage build optimization.
- Managed S3 lifecycle policies, Route 53 DNS, and ACM TLS certificates across application environments.
- Reduced MySQL storage requirements through schema normalization, data archival, and growth-trend analysis, helping avoid unnecessary RDS scaling costs.
- Built Python webhook automation for data synchronization between internal and third-party systems across domains and environments.
- Implemented Google Analytics across client portals to track engagement, conversion funnels, and feature usage for product decisions.
- Centralized application logs and usage metrics in CloudWatch to support monitoring and troubleshooting across AWS services.
- Developed reusable Python scripts for S3, Parameter Store, and RDS/MySQL operations; automated scheduled reports in JSON, Excel, CSV, and Parquet and collected AWS service-usage and cost metrics.

Twellr.ai — AI Hair & Skin Analyzer Platform | *AWS + Azure: EC2 (GPU), Elastic Beanstalk, API Gateway, CloudFront, RDS PostgreSQL, S3, ECR/ACR, Cognito, SSM/Secrets Manager/Key Vault, CloudWatch*

- Designed AWS infrastructure for three product lines using Route 53, CloudFront, API Gateway, Nginx, EC2, and Elastic Beanstalk; deployed Next.js/NestJS and containerized Django services, with development and UAT isolated in separate AWS accounts.
- Deployed Docker-based hair and skin inference workloads on dedicated GPU servers, separating AI services from application workloads; supported shared D2C/B2B inference services protected by signed gateway JWTs.
- Provisioned RDS PostgreSQL, S3 media/model storage, Redis sessions, and RabbitMQ notification queues; restricted S3 delivery through CloudFront Origin Access Control and used Lambda/SES for reporting and transactional email.
- Built per-service GitHub Actions pipelines with Docker builds, versioned ECR images, and SSM Run Command deployments to EC2; validated candidate-container health and retained previous images for rollback; managed Elastic Beanstalk releases.
- Managed secrets using SSM Parameter Store, Secrets Manager, and Azure Key Vault; implemented Cognito OAuth2/OIDC login and RS256 JWT/JWKS service authentication; configured DNS, TLS, and OAC-restricted S3 delivery.
- Deployed the D2C production stack on Azure VMs using ACR, Key Vault, Docker, and GitHub Actions, with DNS cutover pending; supported B2B UAE UAT with tenant subdomains, Azure PostgreSQL, and Blob Storage.
- Centralized application and container logs in per-environment CloudWatch Logs groups; supported reliable releases through health checks, rollback-ready images, and independently deployable service containers.

Supplier Portal | *AWS: EC2, ALB, RDS, S3, Lambda, VPC, Route53, ACM, WAF*

- Designed an Angular frontend on CloudFront/S3 and a Node.js/PM2 backend on EC2 behind an ALB, with RDS MySQL, TLS termination, and a WordPress CRM accessible through the backend.
- Built a VPN-based ETL workflow to extract customer database records, process and synchronize application data, and generate PDF reports for the portal.
- Implemented least-privilege IAM policies for programmatic S3/EC2 access and restrictive security groups to reduce excessive permissions.
- Automated S3 inter-bucket file transfer and cleanup with Lambda event triggers, reducing manual file-operations effort by 70%.
- Diagnosed and resolved a PM2 restart loop causing backend disconnections during UAT, restoring stable service operation.
- Resolved a CORS misconfiguration affecting the Angular frontend and WordPress CRM integration through the Node.js backend.
- Configured ALB, ACM, Route 53, and WAF for TLS termination, custom-domain routing, and web-application protection.

KEY ACHIEVEMENTS

- Reduced monthly AWS spend by approximately 45%, from about $3,000 to $1,500-1,700, through right-sizing and automated scheduling of non-production environments.
- Reduced release time by approximately 80%, from about 4 hours to 45-50 minutes, by automating CI/CD workflows across application and AI/ML services.
- Reduced manual file-operations effort by 70% using Lambda-based S3 transfer and cleanup automation.
- Reduced MySQL storage by 25% through schema optimization, data archival, and growth-trend analysis.

EDUCATION

Bachelor of Engineering — Electronics & Communication Engineering *Aug 2019 – Jun 2023*
Karpagam College of Engineering, Coimbatore, Tamil Nadu